

Appendix A

Diffusion eigenstates and PFGSE amplitude

A serious obstacle in any calculational procedure, for solving the equations either for the spin density $m(\mathbf{r}, t)$ or for the diffusion eigenstates, is the need to satisfy the boundary condition of Eq. (7.3) or Eq. (7.6) at the pore/matrix interface. That is a difficult task for any but the simplest kind of isolated voids microstructure, even when $\rho = 0$. A workable practical method to overcome this difficulty in the latter case is to first extend the partial differential equations (7.2) and (7.5) to the entire system volume V , which includes both the pore space (of volume V_p) and the rock matrix, recasting the problem as a diffusion problem in a composite medium made of two constituents. In order to do that, the diffusion equation must be rewritten in terms of the local chemical potential $\mu(\mathbf{r}, t)$, which is the thermodynamically conjugate variable of the spin density. In contrast with $m(\mathbf{r}, t)$, which even in equilibrium has different values in different constituents, i.e., it is discontinuous across the interface between different constituents, $\mu(\mathbf{r}, t)$ is everywhere a continuous function of $\mathbf{r}$, and has the same constant value everywhere when the system is in equilibrium and there is no net diffusion.

For a system at local equilibrium (i.e., not too far from equilibrium in sufficiently small subvolumes), small changes in $m(\mathbf{r}, t)$ are related to small changes in $\mu(\mathbf{r}, t)$ by means of the local value of the equilibrium thermodynamic derivative $\partial m_{eq}/\partial \mu \equiv m_\mu(\mathbf{r})$, which, like the local value of the diffusion coefficient $D(\mathbf{r})$, has different values in the different constituents. That relation, as well as an expression for $D(\mathbf{r})$, can be written with the help of the characteristic functions of the different constituents—in the present

case these functions are $\theta_p(\mathbf{r})$, which equals 1 if $\mathbf{r}$ is in the pore space and 0 otherwise, and $\theta_m(\mathbf{r}) = 1 - \theta_p(\mathbf{r})$, which equals 1 if $\mathbf{r}$ is in the matrix and 0 otherwise:

$$\begin{aligned}\delta m(\mathbf{r}, t) &= m_\mu(\mathbf{r})\delta\mu(\mathbf{r}, t), \\ m_\mu(\mathbf{r}) &= m_{\mu p}\theta_p(\mathbf{r}) + m_{\mu m}\theta_m(\mathbf{r}), \\ D(\mathbf{r}) &= D_p\theta_p(\mathbf{r}) + D_m\theta_m(\mathbf{r}).\end{aligned}$$

The subscripts p , m denote quantities related to the pore space and the matrix space, respectively. Although both D_m and $m_{\mu m}$ should be taken to vanish, it is sometimes useful, for computational purposes, to allow them to have small positive values. In particular, their relative order of smallness can have a major effect on convergence of the numerical procedure [77].

The case $\rho = 0$

With the help of $D(\mathbf{r})$ and $m_\mu(\mathbf{r})$, which have well defined values everywhere, Eqs. (7.2) and (7.3) can be replaced by a single partial differential equation for $\mu(\mathbf{r}, t)$ when $\rho = 0$

$$m_\mu(\mathbf{r})\frac{\partial\mu}{\partial t} = \nabla \cdot [D(\mathbf{r})m_\mu(\mathbf{r})\nabla\mu]. \quad (\text{A.1})$$

The correct boundary condition at the interface between the two constituents, when D_p and D_m have arbitrary values but $\rho = 0$, is that the normal component of the diffusive flux $D(\mathbf{r})m_\mu(\mathbf{r})\nabla\mu$ is continuous there. That condition is *imposed automatically* by requiring that Eq. (A.1) hold everywhere: in the pore space, in the rock matrix, and also at the pore/matrix interface.

The eigenvalue equations (7.5) and (7.6) are now replaced by the single equation (λ represents the eigenvalue and ϕ_λ the eigenfunction)

$$0 = \lambda m_\mu(\mathbf{r})\phi_\lambda(\mathbf{r}) + \nabla \cdot [D(\mathbf{r})m_\mu(\mathbf{r})\nabla\phi_\lambda(\mathbf{r})]. \quad (\text{A.2})$$

Note that, like $\mu(\mathbf{r}, t)$, $\phi_\lambda(\mathbf{r})$ is everywhere continuous and is defined throughout the entire system volume V , i.e., in the pore space as well as in the matrix space.

If the microstructure is periodic, the eigenfunctions $\phi_\lambda(\mathbf{r})$ are Bloch functions

$$\phi_\lambda(\mathbf{r}) \rightarrow \phi_{n\mathbf{q}}(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}},$$

whose periodic part $\phi_{n\mathbf{q}}(\mathbf{r})$ can be expanded in a Fourier series

$$\phi_{n\mathbf{q}}(\mathbf{r}) = \sum_{\mathbf{g}} \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}) e^{i\mathbf{g}\cdot\mathbf{r}}, \quad \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}) = \frac{1}{V_a} \int_{V_a} dV \phi_{n\mathbf{q}}(\mathbf{r}) e^{-i\mathbf{g}\cdot\mathbf{r}}.$$

Here V_a is the volume of one unit cell, and the sum is over all the vectors $\mathbf{g}$ of the appropriate reciprocal lattice. The eigenvalues now form quasi-continuous bands $\lambda_{n\mathbf{q}}$ and, like the eigenfunctions, they too are characterized by a band index n and a wave vector $\mathbf{q}$, which lies in the first Brillouin zone of reciprocal space. Using the Fourier expansion of $\phi_{n\mathbf{q}}(\mathbf{r})$, the differential equation (A.2) is translated into the following infinite set of linear algebraic equations for the Fourier expansion coefficients $\tilde{\phi}_{n\mathbf{q}}(\mathbf{g})$

$$\begin{aligned} \lambda_{n\mathbf{q}} \sum_{\mathbf{g}'} W_{\mathbf{g}-\mathbf{g}'} \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}') &= D_p \sum_{\mathbf{g}'} \Gamma_{\mathbf{g}\mathbf{g}'}(\mathbf{q}) \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}') \\ \Gamma_{\mathbf{g}\mathbf{g}'}(\mathbf{q}) &= (\mathbf{g} + \mathbf{q}) \cdot (\mathbf{g}' + \mathbf{q}) U_{\mathbf{g}-\mathbf{g}'} \\ U_{\mathbf{g}-\mathbf{g}'} &= \delta_{\mathbf{g}\mathbf{g}'} - u \theta_{\mathbf{g}-\mathbf{g}'}^{(m)} \\ W_{\mathbf{g}-\mathbf{g}'} &= \delta_{\mathbf{g}\mathbf{g}'} - w \theta_{\mathbf{g}-\mathbf{g}'}^{(m)}. \end{aligned} \quad (\text{A.3})$$

Here $\theta_{\mathbf{g}}^{(m)}$ is the $\mathbf{g}$ -Fourier coefficient of $\theta_m(\mathbf{r})$, and is given by

$$\theta_{\mathbf{g}}^{(m)} = \frac{1}{V_a} \int_{V_a} dV \theta_m(\mathbf{r}) e^{-i\mathbf{g}\cdot\mathbf{r}}, \quad (\text{A.4})$$

while w and u are parameters that characterize the heterogeneous thermodynamic and transport properties of the composite:

$$w \equiv 1 - \frac{m_{\mu m}}{m_{\mu p}}, \quad u \equiv 1 - \frac{m_{\mu m} D_m}{m_{\mu p} D_p}.$$

Eq. (A.3) is a *generalized matrix eigenvalue problem*, because of the presence of the $\widehat{W}$ -matrix on the left-hand-side. In order for this problem to yield to standard numerical methods, it is important to keep $\widehat{W}$ positive definite, hence $m_{\mu m}/m_{\mu p}$ should be kept small but greater than 0 in numerical computations (i.e., w close to but less than 1). No such restriction exists regarding the $\widehat{\Gamma}$ -matrix, which is in fact only *positive semi-definite* when $D_m = 0$. If the eigenvectors $\tilde{\phi}_{n\mathbf{q}}(\mathbf{g})$ are normalized as follows ($W(\mathbf{r}) = 1 - w\theta_m(\mathbf{r})$ is the positive periodic function whose Fourier expansion coefficients are $W_{\mathbf{g}}$)

$$\sum_{\mathbf{g}, \mathbf{g}'} \tilde{\phi}_{m\mathbf{q}}^*(\mathbf{g}) W_{\mathbf{g}-\mathbf{g}'} \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}') = \frac{1}{V_a} \int_{V_a} dV W(\mathbf{r}) \phi_{m\mathbf{q}}^*(\mathbf{r}) \phi_{n\mathbf{q}}(\mathbf{r}) = \delta_{mn}, \quad (\text{A.5})$$

we get the following expression for the PFGSE amplitude

$$M(\mathbf{k}, t) = \frac{1}{\phi} \sum_n e^{-\lambda_{n\mathbf{q}} t} |\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}_{\mathbf{k}})|^2, \quad \mathbf{q} = \mathbf{k} - \mathbf{g}_{\mathbf{k}}, \quad (\text{A.6})$$

where $\mathbf{g}_{\mathbf{k}} \equiv \mathbf{k} - \mathbf{q}$ is the reciprocal lattice vector that is closest to $\mathbf{k}$, and

$$\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}) \equiv \frac{1}{V_a} \int_{V_a} dV \theta_p(\mathbf{r}) \phi_{n\mathbf{q}}(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}} = \sum_{\mathbf{g}'} \theta_{\mathbf{g}-\mathbf{g}'}^{(p)} \tilde{\phi}_{n\mathbf{q}}(\mathbf{g}') \quad (\text{A.7})$$

is the $\mathbf{g}$ -Fourier coefficient of $\theta_p(\mathbf{r}) \phi_{n\mathbf{q}}(\mathbf{r})$. The latter function is equal to $\phi_{n\mathbf{q}}(\mathbf{r})$ in the pore space and equal to 0 elsewhere. Truncated versions of Eq. (A.3) (i.e., keeping all the $\mathbf{g}$ -vectors up to some finite upper limit $|\mathbf{g}| \leq g_{max}$) can be solved numerically, using standard methods for matrix eigenvalue problems. Such calculations are described in Refs. [77, 102].

Fig. A.1 shows results for the lowest bands of eigenvalues $\lambda_{n\mathbf{q}}$ in a particular case of a simple cubic spheres array with $\rho = 0$. Because the eigenstates are found by numerical solution of a truncated version of Eq. (A.3), only a finite portion of these eigenvalues can be obtained accurately, i.e., those that are not too large. The contribution of high eigenstates to macroscopic quantities like $M(\mathbf{k}, t)$, $M(t)$, $D(t)$ quickly becomes negligible with increasing t , because they are always weighted by the exponentially decreasing factors $e^{-\lambda_{n\mathbf{q}} t}$. However, for sufficiently short times this method of calculation will always break down.

These calculations lead to results for $D(t)$, using Eq. (7.37), and also for D_e , using the relation [compare with Eq. (7.34)]

$$\lambda_{0\mathbf{q}} \approx D_e \mathbf{q}^2 \text{ for small } \mathbf{q}.$$

Note that, in the correct physical limit $D_m = m_{\mu m} = 0$, w and u are both equal to 1. Thus, $W(\mathbf{r})$ reduces to $\theta_p(\mathbf{r})$, $U_{\mathbf{g}}$ and $W_{\mathbf{g}}$ both reduce to

$$\theta_{\mathbf{g}} = \delta_{\mathbf{g}0} - \theta_{\mathbf{g}}^{(m)} = \frac{1}{V_a} \int_{V_a} dV \theta_p(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}}, \quad (\text{A.8})$$

and the normalization of $\phi_{n\mathbf{q}}$ reduces from Eq. (A.5) to

$$\frac{1}{V_a} \int_{V_a \cap V_p} dV \phi_{m\mathbf{q}}^*(\mathbf{r}) \phi_{n\mathbf{q}}(\mathbf{r}) = \sum_{\mathbf{g}} \tilde{\psi}_{m\mathbf{q}}^*(\mathbf{g}) \tilde{\psi}_{n\mathbf{q}}(\mathbf{g}) = \delta_{mn}. \quad (\text{A.9})$$

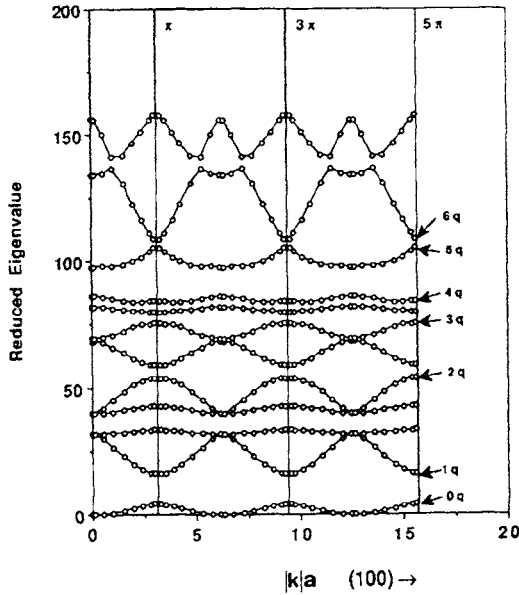


Figure A.1: Results of the 12 lowest reduced (i.e., dimensionless) eigenvalues $\lambda_{n\mathbf{k}}a^2/D_p$, plotted vs. $|\mathbf{k}|a$ along the (100) direction in the one dimensional extended Brillouin zone scheme, for a simple cubic array of identical touching spheres with $\rho = 0$ and $\phi=0.476$, where a is the edge length of the unit cell. The bands which are identified as $0\mathbf{q}$, $1\mathbf{q}$, etc. correspond to the eigenvalues $\lambda_{0\mathbf{q}}$, $\lambda_{1\mathbf{q}}$, etc. The unidentified bands have very small weights and make a negligible contribution to $M(\mathbf{k}, t)$ at all times (after Fig. 2 of Ref. [77]).

The case $\rho \neq 0$

When $\rho \neq 0$, the above approach is not applicable. Instead, one can expand the eigenfunctions in terms of the $\rho = 0$ eigenfunctions for the same microstructure. In the case of a periodic microstructure, both types of eigenfunctions are Bloch functions, and the periodic part of the $\rho \neq 0$ eigenfunctions $\psi_{n\mathbf{q}}(\mathbf{r})$ is expanded in terms of the periodic parts of the $\rho = 0$ eigenfunctions $\phi_{m\mathbf{q}}(\mathbf{r})$:

$$\psi_{n\mathbf{q}} = \sum_m A_m^{(n)}(\mathbf{q}) \phi_{m\mathbf{q}}.$$

The functions $\psi_{n\mathbf{q}}(\mathbf{r})$ are only defined inside the pore space, thus their Fourier coefficients are related to $\tilde{\psi}_{n\mathbf{q}}(\mathbf{g})$ of the $\rho = 0$ problem [see Eq. (A.7)] in the

same fashion

$$\tilde{\psi}_{n\mathbf{q}}(\mathbf{g})\Big|_{\rho \neq 0} = \sum_m A_m^{(n)}(\mathbf{q}) \tilde{\psi}_{m\mathbf{q}}(\mathbf{g})\Big|_{\rho=0}.$$

The expansion coefficients $A_m^{(n)}$ are the solution of the following matrix eigenvalue problem

$$(\mu_{n\mathbf{q}} - \lambda_{m\mathbf{q}})A_m^{(n)}(\mathbf{q}) = \rho \sum_p V_{mp}(\mathbf{q})A_p^{(n)}(\mathbf{q}).$$

Here $\mu_{n\mathbf{q}}$ are the eigenvalues for $\rho \neq 0$ and

$$V_{mp}(\mathbf{q}) \equiv \frac{1}{V_a} \oint_{\partial V_p \cap V_a} dS \phi_{m\mathbf{q}}^*(\mathbf{r}) \phi_{p\mathbf{q}}(\mathbf{r}) \quad (\text{A.10})$$

are *surface overlap integrals* of two $\rho = 0$ eigenfunctions over the pore/matrix interface ∂V_p in one unit cell, with $\phi_{m\mathbf{q}}(\mathbf{r})$ normalized as in Eq. (A.9).

In practice, the matrix elements $V_{nm}(\mathbf{q})$ are evaluated using the Fourier expansion coefficients of $\theta_p(\mathbf{r})\phi_{m\mathbf{q}}(\mathbf{r})$, i.e., the coefficients $\tilde{\psi}_{n\mathbf{q}}(\mathbf{g})$ of the $\rho = 0$ problem [see Eq. (A.7)]

$$V_{nm}(\mathbf{q}) = 4 \sum_{\mathbf{g}} \sum_{\mathbf{g}'} \tilde{\psi}_{n\mathbf{q}}^*(\mathbf{g})\Big|_{\rho=0} K(\mathbf{g} - \mathbf{g}') \tilde{\psi}_{m\mathbf{q}}(\mathbf{g}')\Big|_{\rho=0}, \quad (\text{A.11})$$

where

$$K(\mathbf{g}) \equiv \frac{1}{V_a} \oint_{\partial V_p \cap V_a} dS e^{-i\mathbf{g} \cdot \mathbf{r}}. \quad (\text{A.12})$$

A somewhat subtle but important technical point is the factor 4 which appears in Eq. (A.11): It arises from the fact that the function $\theta_p(\mathbf{r})\phi_{n\mathbf{q}}(\mathbf{r})$ jumps from 0 to $\phi_{n\mathbf{q}}(\mathbf{r})$ when one crosses the interface. Therefore, *at the interface* the Fourier series expansion of that discontinuous function actually converges to 1/2 of the nonzero value of $\phi_{n\mathbf{q}}(\mathbf{r})$ there. The values of $K(\mathbf{g})$ in general depend upon details of the microstructure and periodicity, but $K(\mathbf{0})$ has a simple general value, namely

$$K(\mathbf{0}) = \frac{S_p}{V} = \phi \frac{S_p}{V_p}.$$

It is worth noting that the normalized lowest $\rho = 0$ eigenstate, characterized by $n = 0$ and $\mathbf{q} = \mathbf{0}$, is just

$$\phi_{00}(\mathbf{r}) = \left(\frac{1}{\phi} \right)^{1/2}, \quad (\text{A.13})$$

and that the diagonal matrix element of $\hat{V}$ in that state is

$$V_{00}(0) = \frac{S_p}{V_p}.$$

If the eigenvectors $A_m^{(n)}(\mathbf{q})$ are normalized to 1, then $M(\mathbf{k}, t)$ is given by an expression like Eq. (A.6), but with $\lambda_{n\mathbf{q}}$ replaced by $\mu_{n\mathbf{q}}$, and with $\psi_{n\mathbf{q}}(\mathbf{g})$ now representing the Fourier expansion coefficients of the $\rho \neq 0$ eigenfunctions, namely

$$\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}) \equiv \frac{1}{V_a} \int_{V_p \cap V_a} dV \psi_{n\mathbf{q}}(\mathbf{r})|_{\rho \neq 0} e^{-i\mathbf{g} \cdot \mathbf{r}}.$$

That explains and justifies Eq. (7.63), which can be used both for $\rho = 0$ and $\rho \neq 0$, if $\mu_{n\mathbf{q}}$ and $\psi_{n\mathbf{q}}(\mathbf{g})$ are chosen appropriately.

Fig. A.2 shows results for the same microstructure as Fig. A.1, but with two nonzero values of ρ . As in the $\rho = 0$ case, here too the high eigenstates will not be calculated reliably because of the matrix truncation. Consequently the calculation of $M(\mathbf{k}, t)$, $M(t)$, $D(t)$ breaks down for sufficiently short t .

As was the case when $\rho = 0$, the low $\mathbf{q}$ eigenvalues of the lowest band provide the value of D_e

$$\mu_{0\mathbf{q}} = \mu_{00} + D_e \mathbf{q}^2 + \mathcal{O}(\mathbf{q}^4),$$

except that μ_{00} is nonzero, and the value of D_e depends on ρ .

The time dependent diffusion coefficient

Using Eq. (7.37), the small $\mathbf{k}$ behavior of $M(\mathbf{k}, t)$ again leads to results for the time dependent diffusion coefficient $D(t)$. However, there is now a somewhat subtle dependence of $D(t)$ on the precise way it is defined. The most natural physical definition of $D(t)$ is based on the mean square displacement of a diffusing particle, and is motivated by the result for a uniform fluid, given by Eq. (7.24):

$$\langle [\mathbf{r}(t) - \mathbf{r}(0)]^2 \rangle \equiv 6D(t)t, \quad (\text{A.14})$$

where the average is restricted to diffusing particles that survive (i.e., spins that did not depolarize) at time t . This average can be calculated as follows:

$$\langle [\mathbf{r}(t) - \mathbf{r}(0)]^2 \rangle = \frac{1}{V_p} \int_{V_p} dV' \frac{\int_{V_p} dV (\mathbf{r} - \mathbf{r}')^2 G(\mathbf{r}, \mathbf{r}'|t)}{\int_{V_p} dV G(\mathbf{r}, \mathbf{r}'|t)}, \quad (\text{A.15})$$

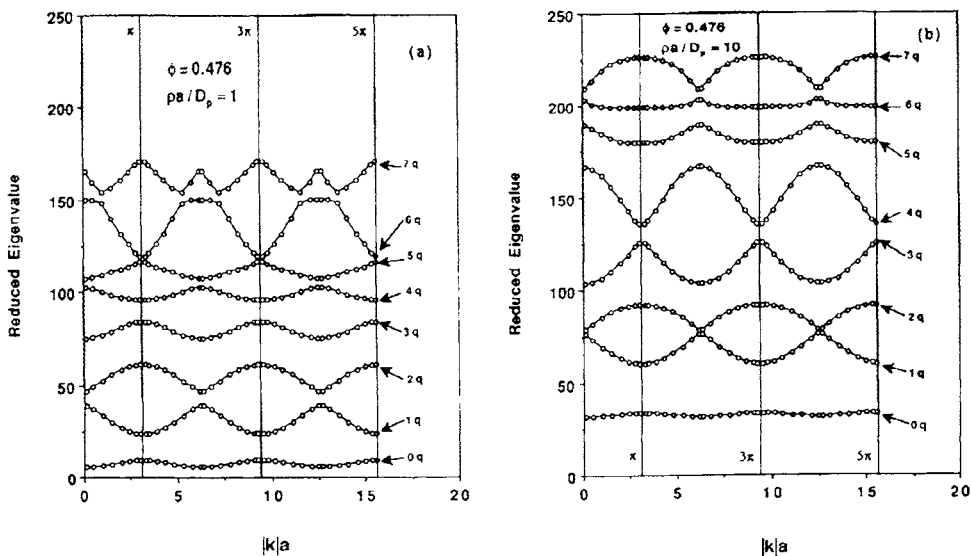


Figure A.2: Results for the 7 lowest reduced (i.e., dimensionless) eigenvalues $\mu_{n\mathbf{k}}a^2/D_p$, plotted vs. $|\mathbf{k}|a$ along the (100) direction in the one dimensional extended zone scheme, for a simple cubic array of identical touching spheres ($\phi=0.4764$) (a) for $\rho a/D_p = 1$ (b) for $\rho a/D_p = 10$. The bands which are identified as $0\mathbf{q}$, $1\mathbf{q}$, etc. correspond to the eigenvalues $\mu_{0\mathbf{q}}$, $\mu_{1\mathbf{q}}$, etc. (after Fig. 2 of Ref. [18]).

where the integral $\int_{V_p} dV G(\mathbf{r}, \mathbf{r}'|t)$ is the probability of survival of the particle that was originally at $\mathbf{r}'$. Only when the decay of $M(t)$ can be neglected, can this integral be replaced by 1, and only then can we write

$$6D(t)t = - \frac{1}{M(0, t)} \left. \frac{\partial^2 M(\mathbf{k}, t)}{\partial \mathbf{k} \cdot \partial \mathbf{k}} \right|_{\mathbf{k}=0}, \quad (\text{A.16})$$

which is equivalent to Eq. (7.37). Although the uniform exponential decay rate $1/T_{2B} + \nu_0$ can be factored out of $M(\mathbf{k}, t)$, the same cannot be done for the ρ dependence of the decay. Thus, when ρ is not small, the $D(t)$ calculated from Eq. (7.37) or Eq. (A.16) will differ from the $D(t)$ of Eq. (A.14). Since Eq. (A.16) is based upon a macroscopically measurable quantity, namely the PFGSE amplitude $M(\mathbf{k}, t)$, it is sometimes preferable to use that equation to define $D(t)$, instead of the more natural intuitive physical definition implied by Eqs. (A.14) and (A.15).

Decay of the total polarization

The $\mathbf{q} = 0$, $n \neq 0$ eigenvalues have non-trivial values for $\rho \neq 0$, as well as for $\rho = 0$. However, when $\rho = 0$, they make no contribution to the decay rate of the total spin polarization $M(t) = M(0)M(\mathbf{k} = 0, t)$, which is constant in time (apart from the constant, microstructure independent decay rate $1/T_{2B} + \nu_0$, which is ignored here). This is due to the fact that the $\mathbf{q} = 0$, $\mathbf{g} = 0$, $n \neq 0$ Fourier expansion coefficients all vanish

$$\tilde{\psi}_{n0}(\mathbf{0}) = 0 \text{ for } n \neq 0, \text{ when } \rho = 0,$$

because $\phi_{n0}(\mathbf{r})$ is orthogonal to $\phi_{00}(\mathbf{r})$, which is a constant.

By contrast, when $\rho \neq 0$, the analogous coefficients do not vanish. Therefore the eigenvalues μ_{n0} provide an entire spectrum of non-trivial decay rates for $M(t)$ via

$$M(\mathbf{0}, t) = \frac{1}{\phi} \sum_n e^{-\mu_{n0}t} |\tilde{\psi}_{n0}(\mathbf{0})|^2.$$